

Model Inputs, Ranked — What to Feed a Price/Outlook Model

Research note, not a feature spec. Written to inform prompts 11+ and any future prediction work. Measured against this repo on 2026-08-30, with Kamada 0001567529 (381 filings, 2016-01-06 → 2026-08-20) as the **development fixture** — one company for practice, with the corpus scaling to as many issuers as the work needs.

0. Two caveats that outrank every input below

A. The target must be defined first. "Predict the stock" is not a target. Next-day reaction to a 6-K, 90-day post-earnings drift, and 12-month total return have almost disjoint top-fives. This note assumes **short-horizon reaction and drift around company events** (1–90 days), because that is what the timeline is built around. State it explicitly before building, or the feature set will be right for a question nobody asked.

B. Point-in-time discipline is the difference between a model and a fantasy. Consensus estimates get restated, financials get amended, tickers get reassigned. Every input below must be stored **as it was known on the date it was known** — exactly what prompts 8–9's accession-keyed artifacts already do, and exactly what a naive "fetch current consensus" call would destroy. This is the most common way a backtest lies to its author, and it is much cheaper to design in now than to retrofit across a large corpus.

The single-company corpus is a *practice fixture*, not a constraint on the design. What follows assumes breadth arrives; §2 is about building so that it can.

1. The ranking

Ranked by **(evidence strength × availability) ÷ engineering cost**. "Have it" = in the corpus or DB today. **X-sec** marks inputs that only become meaningful once the corpus holds many companies.

#	Input	Evidence	X-sec	Have it?	Cost
1	Earnings surprise — actual vs consensus (revenue, EPS)	Strong	—	Actuals ✓ / consensus ✕	Low once consensus exists
2	Sector & peer classification (SIC, filer status) — the enabler for every relative feature	Foundational	✓	✓ free, on disk	Trivial
3	Forward guidance — issued, raised, cut, withdrawn	Strong	—	In EX-99 prose, unextracted	Medium
4	Price & volume history — returns, volatility, liquidity, abnormal volume	Strong (the <i>label</i> source)	—	✓ shipped	Done
5	Market & sector factor returns — to turn raw returns into <i>abnormal</i> returns	Strong	✓	✕	Low
6	Estimate revisions — direction, velocity, and breadth across peers	Strong	partly	✕	Medium
7	Event type & timing — filing category, cadence, off-cycle 6-Ks	Moderate	—	✓ shipped	Done
8	Balance-sheet durability — cash, burn, runway, debt	Strong for small/mid-cap	—	✓ extracted	Low

9	Sector-conditional domain events — clinical/FDA for pharma, rig counts for energy, ...	Strong <i>within</i> a sector	—	✗	Medium per sector
10	Short interest	Moderate	—	✗	Low
11	Segment / product mix shifts	Moderate	—	✓ partially	Low
12	Insider open-market purchases (code P only)	Moderate, asymmetric	—	△ see §5	Medium
13	Analyst rating / PT changes	Weak–moderate	—	✗	Low
14	Institutional ownership (13F)	Weak (45-day lag)	✓	✗	Medium
15	News sentiment	Weak, expensive to do well	—	✗	High
16	Social / retail chatter	Noise at this market cap	—	✗	High

2. What breadth unlocks — and what to build now so it works

Four things do not exist at $n=1$ and are the main reason to design for scale early.

2.1 Relative beats absolute

A +12.6% revenue surprise means nothing in isolation. Its **percentile within its sector** in that quarter is the feature that carries signal. The same is true of margin change, revision direction, and every ratio. Once the corpus has peers, most raw features should be replaced by their cross-sectional rank — which requires storing the raw value per company per period, which is what prompts 8–9 already do. **No change needed; just don't collapse history into "latest".**

2.2 Peer grouping is already free — measured

`submissions.json` already carries everything needed to group companies, with no external classification vendor:

```
sic 2834
sicDescription Pharmaceutical Preparations
ownerOrg 03 Life Sciences
category Accelerated filer
fiscalYearEnd 1231
exchanges ['Nasdaq']
```

SIC gives the peer set, `ownerOrg` gives a coarser sector, `fiscalYearEnd` is required to align periods across companies (comparing a June-FY company to a December-FY one on "FY2025" is a real and easy mistake), and `category` (filer status) drives filing deadlines. This is why classification ranks **#2**: it costs nothing and every relative feature depends on it.

2.3 Abnormal return, not raw return

With many companies the label should be the **excess** return over market and sector on the event window. A 6-K that "moved the stock +4%" on a day the sector rose 4% carries no information. This needs a market index and a sector aggregate — the latter computable from the corpus itself once enough peers exist, so it is cheap. Skipping this makes every label partly a proxy for beta.

2.4 Survivorship is the trap that only appears at scale

If the universe is built from *today's* listed companies, every delisted, acquired, and bankrupt issuer is silently excluded, and the backtest inherits a strong upward bias. EDGAR retains filings for companies that no longer trade, so the fix is available: **build the fetch list from historical constituents, not current ones**, and keep companies in the corpus after they stop trading. This is invisible at n=1 and expensive to correct after the corpus is large.

3. Why the top of the list looks like that

#1 Earnings surprise — and why it reframes prompt 11

The most robustly documented equity anomaly is **post-earnings-announcement drift**: prices keep moving in the direction of a surprise for weeks after the print. Surprise is a *difference*, so it needs two terms:

- **Actual** — already extracted, verified, point-in-time, accession-keyed (prompts 8–9).
- **Consensus** — **not held anywhere in this system**.

That is the gap, and it means prompt 11's consensus panel is not a nice-to-have tab: **it is the missing denominator of the highest-value input**. Finnhub/FMP consensus is the cheapest path to it — a far stronger argument for that prompt's Tier 2 than "show the user a rating split."

The retrofit question splits in two, and an earlier draft of this note got it wrong by treating both halves the same:

- **Historical EPS surprise is retrofittable**. Vendors return, per past quarter, the estimate *as it stood at that announcement* alongside the actual. One call per company recovers the whole history — no prior snapshotting required. This is the #1 feature, and it is cheaper than previously stated.
- **Forward estimates and revision velocity (#6) are not**. Those endpoints return only *today's* view. Every week without a snapshot is a week of revision history that cannot be recovered later. This is the genuinely perishable half, and the reason to start snapshotting before the corpus grows.

Both halves still need an `as_of` date stored per row — see §4 for why that matters more than it first appears.

#3 Guidance — often the bigger mover than the print

Forward guidance ("raising full-year revenue guidance to \$158–162 million") frequently moves the stock more than the quarter itself. Measured: that exact sentence sits in an EX-99 exhibit in this corpus, unextracted — the same exhibits prompt 9's adjudicator already pulls unit-explicit prose from. Guidance issued / raised / cut / withdrawn is a small extraction with high value, and an LLM is genuinely good at it.

#9 Sector-conditional domain events — reranked

In the previous version this sat at #5 as "clinical/regulatory", because the fixture company is a plasma-derived pharma business where a Phase 3 readout can reprice the equity more than a decade of earnings. That is still true **for pharma**, and ClinicalTrials.gov and openFDA are free and structured.

But in a multi-sector corpus it is one of a family of sector-specific feeds, each valuable only within its sector. It moves to #9 as a *class*, while staying near the top for any healthcare-heavy universe. Which sector to build first should follow the corpus composition — if the universe is mostly pharma, build it early.

4. Consensus — where it comes from, and why it is not objective

Sourcing, measured. Tested 2026-08-31:

Source	Gives	Depth	Cost	Status
Finnhub /stock/earnings	EPS actual, estimate, surprise, surprisePercent	Deep history	Free key, per-minute limit	Fields confirmed from the official Go client's EarningResult model

FMP <i>analyst-estimates</i> / earnings	Revenue + EPS estimates	Historical + forward	Free key, 250 calls/day	Documented
Alpha Vantage <i>EARNINGS</i>	EPS <i>reported</i> EPS, <i>estimated</i> EPS, <i>surprise</i>	122 quarters for IBM	Free key, very low daily cap	Tested live , HTTP 200
Nasdaq /api/company/{sym}/earnings-surprise	EPS actual + consensus	4 quarters only	No key	Tested live , HTTP 200
Nasdaq /revenue	—	—	—	Tested: "Data not available"

Revenue consensus is the binding gap. Finnhub's free earnings endpoint is EPS-only and Nasdaq has no revenue; FMP is the only free route to revenue estimates, and its 250/day cap is what decides feasibility. A watchlist of ~200 names fits in a day; a 1,000-name universe needs roughly 17 minutes on Finnhub for EPS but **four days** on FMP for revenue — which is where a paid tier stops being optional.

Consensus is a vendor construct, not a measured quantity. Same company, same quarters, two vendors (IBM, 2026-08-31):

Quarter	Actual (both agree)	Alpha Vantage est.	Nasdaq est.
2026-06-30	2.93	2.93	2.93
2026-03-31	1.91	1.81	1.81
2025-12-31	4.52	4.29	4.33
2025-09-30	2.65	2.45	2.44

The **actuals agree exactly**; the **consensus disagrees in half the quarters**. On Dec-2025 the implied surprise is 0.23 against 0.19 — a ~20% relative difference in the modelled feature, arising purely from vendor choice.

It cannot be otherwise. Consensus is a statistic over a population each vendor defines differently: the **contributor panel** (which firms submit), the **staleness rule** (how recently an estimate must have been revised to count), **basis normalization** (vendors restate contributors onto a house definition of EPS — GAAP vs street, stock comp, intangible amortization, one-offs), **mean vs median and outlier trimming, as-of timing** (consensus drifts daily), **share count and FX** (diluted assumptions, ADR ratios, reporting currency — live issues for an Israeli IFRS filer), and **fiscal alignment**.

And vendors **revise history**: I/B/E/S analyst records have been documented as altered retroactively (Ljungqvist, Malloy & Marston, *Rewriting History*, Journal of Finance 2009). So "historical consensus" pulled today may not equal what was on screen then — a direct threat to §0 caveat B that no amount of local discipline can fix.

The rules this implies

1. **Surprise = vendor actual – vendor estimate.** Both sides on one basis, internally consistent. This is the number to model.
2. **The SEC-extracted actual is a basis check, not the numerator.** This corrects an earlier draft. IBM's *reported*EPS of 2.93 is *street* EPS, not GAAP; computing *our* GAAP *actual* – *vendor street estimate* manufactures a surprise out of a basis mismatch. Use the comparison to *classify* the vendor: if vendor actual ≈ our extracted actual, the consensus is on a reported basis; if it diverges, the vendor is on street basis. Either is usable — mixing them is not.
3. **Match on (period end and duration).** Measured on the fixture company: joining on period end alone compared the 20-F's full-year EPS of 0.35 against a Q4 consensus of 0.09, manufacturing a

~289% surprise from nothing. `financials.json` already carries `period.duration` (`P3M/P1Y`), so this costs nothing to get right.

4. **One vendor per feature, never mixed.** The cross-vendor spread above is noise that would sit on top of an already-weak signal.
5. **Prefer scaled surprise (SUE-style)** — surprise divided by its own historical standard deviation for that company. A systematic vendor basis offset largely cancels, making the feature robust to exactly this ambiguity.
6. **Store `vendor` and `as_of` on every row**, so a later vendor switch is detectable rather than silently corrupting a backtest.

5. Insider transactions — the measured reality

The evidence is real but **narrower than the folklore**: open-market insider *purchases* (Form 4 code **P**) carry modest predictive power; *sales* are mostly uninformative (diversification, tax, 10b5-1 scheduling); grants and vesting carry essentially none.

Measured on the fixture company:

Form 4 filings	26 (all parse cleanly)
Form 3 filings	20
Transactions extracted	227
Transaction codes present	A 118, D 109
Code P (open-market purchase)	0
Code S (open-market sale)	0
Transactions carrying a price	218 of 227
Insider history depth	2026-03-13 → 2026-08-06 (≈5 months) vs 10.6 years of filings

1. **Every insider event currently on disk is compensation mechanics** — awards and dispositions to the issuer, not conviction trades. Feeding these to a model adds noise. An implementation must filter to `P/S` and treat `A/D/F/M` as equity-comp events (useful for dilution accounting, not direction).
2. **The history is 5 months deep** because the EDGAR fetch only pulled recent Forms 3/4 — the filings exist upstream and were never requested. `submissions.json` even flags `insiderTransactionForIssuerExists`, so the backfill target is knowable per company.

Worth building — free, structured XML, already half-parsed, and the `ownership.xml` schema is stable — but **#12, not top-five**, and with the code filter stated up front. Across many companies the `P` population will be thin but non-zero, which is exactly the case where breadth converts a useless single-name feature into a usable one.

6. Scaling constraints — measured, and they bind

Constraint	Measured	Implication
SEC fair-access limit	10 requests/second , enforced per IP across all EDGAR domains (<code>www</code> , <code>data</code> , <code>efits</code>), regardless of how many machines you run	The fetcher must be globally rate-limited, not per-worker. Exceeding it throttles the IP until the rate stays down for 10 minutes

Disk per company	647 MB, 4,422 files	~650 GB per 1,000 companies as currently fetched
...of which .txt	386 MB (60%)	The full-submission {accession}.txt files duplicate documents already stored individually. Dropping them cuts the corpus ~60%
...of which .jpg	78 MB (12%)	Investor-deck images; no modelling value
Company Facts	1.1 MB, one fetch per CIK	Already correct — cached per company, not per accession (prompt 9)
Price rows	2,673 per symbol (~10.6y daily)	Trivial; ~2.7M rows at 1,000 companies

Pruning .txt and .jpg takes a company from 647 MB to roughly **180 MB** — 650 GB down to 180 GB at a thousand names. Worth deciding **before** a large fetch, since re-fetching to prune is far more expensive than not storing it.

7. What I would build next, in order

1. **Consensus** (prompt 11 Tier 2) — the highest-leverage single addition. Two jobs, not one: a **backfill** of historical EPS surprise (retrofittable, one call per company) and a **recurring snapshot** of forward estimates (perishable — start it first). Pick one vendor and record it per row (§4).
2. **Corpus expansion** — with the fetch list built from *historical* constituents (§2.4) and .txt/.jpg pruned (§6). Both decisions are cheap now and costly later.
3. **Guidance extraction from EX-99** — reuses prompt 9's prose-passage machinery.
4. **Peer-relative feature layer** — SIC grouping is already on disk; convert absolute metrics to sector-relative ranks, with *fiscalYearEnd* alignment.
5. **Abnormal-return labels** — market/sector adjustment on event windows.
6. **Form 4 parser with p/s filtering + backfill.**
7. **Clinical/FDA calendar** — early if the universe stays healthcare-weighted.

Deliberately *not* near-term: news sentiment, social, 13F — expensive per unit of signal, and two of the three are laggy by construction.

8. Honest limits

- Published anomalies decay after publication; PEAD and insider-purchase effects are weaker now than in the studies that established them.
- The #1 feature rests on a number that is **defined, not measured** (§4). Results will shift with vendor choice, and a vendor's retroactive edits to history are outside your control. Report which vendor a result came from.
- Every input here is weak individually. Combining weak signals across a broad cross-section is what works — which is the real argument for corpus breadth, over and above having more rows.
- Backtest results on a corpus you assembled yourself will be optimistic. Reserve a holdout period from the start and do not look at it.
- None of this is investment advice. The defensible product is a research aid — a way to read filings and spot deviations faster — not a trading signal.

References

- Consensus sourcing — [Finnhub EarningResult model](#), [FMP Financial Estimates](#), [FMP limits](#)
- Vendors revising history — Ljungqvist, Malloy & Marston, "Rewriting History", *Journal of Finance* 64(4), 2009
- SEC fair access / rate limits — [Accessing EDGAR Data](#), [new rate control limits](#)
- Prior art in this repo: `prompt_8_extracting_info_from_filings-hld.md` (point-in-time sidecar artifacts), `prompt_9_extra_source_for_financial_reports-hld.md` (provider cache + coverage row), `prompt_11_company_analysts-hld.md` (consensus source tiers)